GR00T N1.7 policy

Visual observation

Language instruction

"Pick up the bottle"

Tokenizer

Vision encoder

Token embedding

Projector

Fixed 4-bit

SACQ scope

Language backbone
112 linear layers

DiT action head
226 linear layers

Action chunk

1 Structure-aware codebook fitting

BF16 weights

Indices a

Codebook $\mathcal{C}$

Weighted k-means

4D codewords

Role-specific metric

Query example: preserve attention logits

Key-channel importance

4D input correlations

$B_{r(v),r(v)}$ × $A_{I(v),I(v)}$ = $G(v)$

$$\min_{\mathcal{C},a} \sum_{v} (v - c_{a(v)})^{\top} G(v) (v - c_{a(v)})$$

2 Action-aware mixed-bit allocation

Candidate states

GPTQ-4b → GPTQ-3b → VC-2.75b → ... → VC-1b

Layer-wise action sensitivity

Current allocation

Lower precision of layer ℓ

Δ_ℓ

Action-error change from a one-step precision reduction

Greedy layer selection

$$\text{Score}_\ell = (\Delta_\ell)/(\text{bits saved}_\ell)$$

Layer 1

Layer 2 → Lowest score

Layer 3

Mixed-bit assignment

VC-2b | VC-1b | VC-1.5b | GPTQ-3b

Layer 1 | Layer 2 | Layer 3 | Layer 4

3 Post-quantization refinement

Index reassignment

Update indices with $\mathcal{C}$, s and bit allocation fixed

Before

After

H_ℓ

U_ℓ

Input activation statistics

Cholesky factor

$$\widetilde{H}_\ell^{-1} = U_\ell^{\top} U_\ell$$

Action-level calibration

Update $\mathcal{C}$ and s with indices and bit allocation fixed

BF16 policy

$\mathbf{a}_{\text{BF16}}$

Quantized policy

$\mathbf{a}_Q$

Action MSE

Optimize $\mathcal{C}$, s

$$\min_{\mathcal{C},s} \mathbb{E}_x \left[\text{MSE}\left(\mathbf{a}_Q(x), \mathbf{a}_{\text{BF16}}(x)\right) \right]$$